

DBMS Unit 3 Notes: Relational Schema & Normalization

DEPARTMENT OF COMPUTER ENGINEERING - COURSE: DATABASE MANAGEMENT SYSTEMS

Unit 3: Relational Database Design and Normalization Theory

1. Functional Dependencies:

A functional dependency $X \rightarrow Y$ specifies that if two tuples agree on attributes X, they must also agree on attributes Y.

2. Normal Forms (1NF, 2NF, 3NF, BCNF):

First Normal Form eliminates repeating groups and ensures atomic attributes. Second Normal Form eliminates partial functional dependencies. Third Normal Form eliminates transitive dependencies.

3. ACID Properties in Transaction Processing:

Atomicity, Consistency, Isolation, and Durability ensure reliable transaction execution and database integrity.